SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:50**

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I Christopher *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

APPLICATION- BASED QUESTIONS

1. **Scenario: Event Communication Security**
A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.

   a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
   b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. **Scenario: Intercepted Suspicious Message**
A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

   a. Deduce the most likely original message.
   b. Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

3. **Scenario: Secure Messaging System**
A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.

   a. Demonstrate how the encryption process transforms the message step-by-step.
   b. Explain how the repeating keyword influences each character of the ciphertext.

4. **Scenario: Suspicious Digital Asset Investigation**
During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

   a. Analyze how investigators could detect the presence of hidden data within the image.
   b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. **Scenario: Legacy Encryption in Corporate Communication**
A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEEFEAOCAIVDEN".

a. Determine the most probable original message by reversing the encryption process.
b. Critically evaluate why such a method is vulnerable in modern communication systems.
c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools

Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

CO 1 - Assessment Tool 1

1) Event communication Security - Caesar cipher.

a) Reconstruct the encrypted message using a shift of 4.

Original message : MEET AT NOON.

Using Ceasar cipher, shift each letter 4 positions forward in the alphabet.

M → Q
E → I
E → I
T → X
A → E

T → X
N → R
O → S
O → S
N → R.

Encrypted message :- QIIX EX RSSR.

b) How can an authorized member retrieve the original message?

The authorized member (decrypts) decrypts the message by shifting each letter 4 positions backward in the alphabet.

Example :- Q → M ; I → E ; I → E ; X → T.

Thus, the original message is recovered.

Original message :- MEET AT NOON.

encrypted message :- QIIX EX RSSR.

2) Intercepted suspicious Message - Substitution cipher.

a) Deduce the most likely original message.

Intercepted message :- GSRH RH Z HVXING.

using the general monoalphabetic cipher;

where A → Z, B → Y, C → X, and so on.

so,

original message :- THIS IS SECRET.

b) Justify your answer by analyzing repeating patterns, letter structures, (or) known cipher characteristics.

→ The message uses a consistent letter - substitution pattern (general monoalphabetic substitution cipher).

→ The substitution technique replaces letter with its reverse - alphabet counterpart.

→ For example :- G → T, S → H

R → I, H → S.

→ The decoded words form a meaningful English sentence.

Eg:- THIS IS SECRET.

→ Therefore, the message was most likely encrypted with monoalphabetic substitution cipher.

3) Secure Messaging System – Vigenere cipher.

a) Demostrate the encryption process step-by-step

plaintext :- HELLO.

key : KEY.

7	4	11	11	14
H	E	L	L	O
K	E	Y	K	E
10	4	24	10	4

Encryption :-

$$C_i = P_i + k_i \pmod{26}$$

$C_H = 7 + 10 \bmod 26$

$= 17$ (R)

$C_E = 4 + 4 \bmod 26$

$= 8$ (I)

$C_L = 11 + 24 \bmod 26$

$= 35 \bmod 26 = 9$ (J)

$C_L = 11 + 10 \bmod 26$

$= 21$ (V)

$C_O = 14 + 4 \bmod 26$

$= 18$ (S).

Cipher text :- RIJVS.

b) How does the repeating keyword influence each character?

→ Each plaintext letter is combined with corresponding key letter.

→ Different key letters produce different shifts.

→ The keyword repeats when the message is longer than the key.

→ This makes the cipher more complex than ceasar cipher.

4) Suspicious Digital Asset Investigation - Steganography:

a) How can investigators detect hidden data in an image?

→ Examine the image's metadata for suspicious information.
→ Compare the file size with that of similar images.
→ Analyze pixel values and least significant bits (LSB's).
→ Use steganography detection tools to identify usual patterns.
→ Extract suspected hidden data using appropriate forensic tools.

b) Recommend two techniques to strengthen security.

1) Encryption before embedding:-

Encrypt the secret message before hiding it in the image, so extracted data remains unreadable without the key.

2) Randomized embedding:- Use a secret key to select pixel locations instead of embedding data in predictable positions, making detection and extraction more difficult.

5) Legacy Encryption in corporate communication - Transposition Cipher.

Recieved message :- WERRLTEERDSOEEFEAOCAIVDEN.

a) Determine the original message by reversing the encryption.

The ciphertext appears due to use a columnar transposition cipher. However, the exact arrangement of the three levels is not specified clearly though to determine a unique decryption with certainity.

A common transposition example is :-

→ Original message :- WE ARE DECIDED FLEE AT ONCE.

→ Remove spaces :- WEAREDECIDEDFLEEATONCE.

→ Rearrange the letters according to the specified transposition pattern to produce the ciphertext.

b) Why is this method vulnerable in modern communication system ?

→ It rearranges letters without changing them.

→ Letter frequencies remain the same as in the plaintext.

→ Repeated patterns can help attackers identify the message.

→ It does not provide modern cryptographic security.

→ It is vulnerable to cryptoanalysis, especially when the arrangement is known or guessed.

c) propose one practical enhancement.

Replace the transposition cipher with AES (Advanced Encryption Standard) using a secure mode such as AES-

GCM. It provides strong confidentiality and also detects unauthorized modification of the encrypted message.